

Ayberk Sadiç

Robotics Engineer

PhD candidate · Institute of Science Tokyo · Tokyo, Japan

ayberksadic@gmail.com · ayberksadic.com · [LinkedIn](#)

Robotics engineer working across mechanical design, embedded electronics, software, and assistive robotics. Experience leading multidisciplinary development from mechanisms and prototypes to integrated systems.

EXPERIENCE

Robotics Engineer · Yukai Engineering

Aug 2025 — Present

Tokyo, Japan

- Mechanical design, embedded software, and multilayer PCB development for robotic products.
- Sensor, actuator, and Linux platform integration; hardware bring-up, debugging, and validation.
- Development of a conversational AI assistant platform and backend infrastructure.

Founder & Systems Architect · Mekadermis

Jan 2018 — Feb 2024

Istanbul-based startup focused on robotics and automation

- Delivered customized robotics and automation solutions to industrial clients.
- Led mechanism design, kinematics and inverse dynamics analysis, prototyping, and testing.
- Developed embedded hardware, firmware, actuator drivers, sensor interfaces, and control systems.
- Managed multidisciplinary teams, product development cycles, technical roadmaps, and budgets.

Research & Teaching Assistant · Koç University

Aug 2016 — Jan 2018

Robotics & Mechatronics Laboratory · Istanbul, Turkey

Research on robotics and human-machine interaction; teaching in control systems design and bachelor's thesis courses.

EDUCATION

PhD candidate, Mechanical Engineering

Apr 2024 — Mar 2027 (expected)

Institute of Science Tokyo · Mechanical Systems Design Laboratory

Master of Business Administration

Sep 2019 — Mar 2023

Galatasaray University · Research on financial anxiety, financial literacy, and mental accounting

BSc, Mechanical Engineering

Sep 2011 — Jun 2016

Middle East Technical University

TECHNICAL EXPERTISE

Mechanical & robotics: Mechanism design, kinematics and dynamics, CAD/CAE, control systems, Autodesk Fusion / Inventor, SolidWorks, MATLAB / Simulink.

Embedded & electronics: C/C++, STM32, ESP32, ARM Cortex-M, FreeRTOS, multilayer PCB design, I²C, SPI, UART, CAN, hardware testing and debugging.

Software & research: Python, ROS1/2, embedded Linux, OpenCV, Git, Docker, AnyBody Modeling System, motion capture, EMG, biomechanical analysis.

SELECTED PUBLICATIONS

Design and Task Space Evaluation of 3-URS Parallel Human Neck Assistance Robot. A. Sadiç & Y. Sugahara. ROMANSY 26 – Robot Design, Dynamics and Control, Springer, 2026, pp. 44–54. [doi:10.1007/978-3-032-34511-0_5](https://doi.org/10.1007/978-3-032-34511-0_5)

A K-Fold Cross-Validated Regression Model for Cervical Spine Pose Estimation Trained on X-Ray Images. A. Sadiç & Y. Sugahara. IEEE ROBIO, 2025, pp. 49–54. [doi:10.1109/ROBIO66223.2025.11376001](https://doi.org/10.1109/ROBIO66223.2025.11376001)

LANGUAGES

Turkish: native · English: TOEFL iBT 109/120, TOEIC L&R 980/990 · Japanese: conversational